
BY 506: Advanced Immunology

Course Code	BY 506
Course Name	Advanced Immunology
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	3rd and 4th year UG and PG (MS/MTech, PhD)
Distribution	Elective
Approval	10th Senate (Jun 2016)

Course Contents

- **Introduction**
Cells of immune system; innate and acquired immunity; primary and secondary lymphoid organs; antigens: chemical and molecular nature; haptens; adjuvants; types of immune responses; theory of clonal selection [8 Lectures]
- **IMMUNE RESPONSES** Development, maturation, activation and differentiation of T-cells and B-cells; TCR; antibodies: structure and functions; antibodies: genes and generation of diversity; antigen-antibody reactions; monoclonal antibodies: principles and applications; antigen presenting cells; major histocompatibility complex; antigen processing and presentation; regulation of T-cell and B-cell responses. [16 Lectures]
- **INFECTION AND IMMUNITY**
Injury and inflammation; immune responses to infections: immunity to viruses, bacteria, fungi and parasites; cytokines; complement; immunosuppression, tolerance; allergy and hypersensitivity; Immunodeficiencies; resistance and immunization; Vaccines. [15 Lectures]
- **Immuno-technology**
Autoimmunity, Autoimmune disorders and diagnosis [4 Lectures]

Text Books

1. Kuby J, **Immunology**, 7th edition, MacMillan press
2. Janeway, Charles A., et al., **Immunobiology: The Immune System in Health and Disease**, Garland Science.
3. Peter Delves, Seamus Martin, Dennis Burton, Ivan Roitt, **Roitt's Essential Immunology (Essentials)**, Wiley-Blackwell, 2006.

References

1. William E Paul, **Fundamental Immunology**, Lippincott Williams & Wilkins, 2008 .
2. Gerald B. Pier, Jeffrey B. Lyczak, Lee M. Wetzler, **Immunology, Infection, and Immunity**, ASM Press, 2004.
3. Ian Todd, Gavin Spickett, **Lecture Notes: Immunology**, 5th Edition, Wiley -Blackwell, 2005.
4. Richard Coico, Geoffrey Sunshine, **Immunology: A Short Course**, Wiley-Blackwell, 2009

Articles

Relevant articles and scientific papers will be provided during class room teaching.